

Bernstein Energy & Power: Powering Data-Centers beyond American shores...

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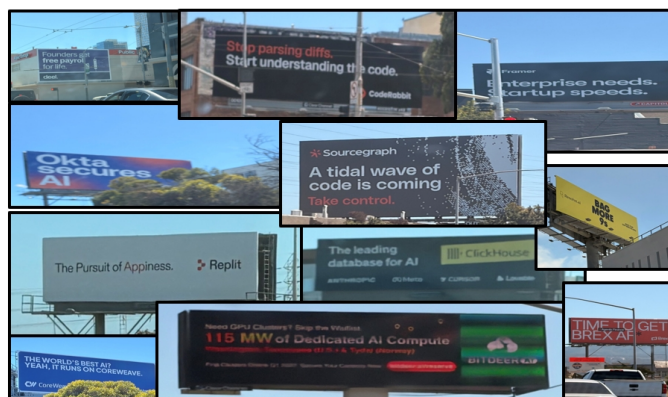
By Nikhil Nigania and Uma Menon

THE US BOOM - WHAT IS IT LIKE TO BE IN THE BAY-AREA AT THIS POINT?

I met ~20 individuals across big tech / frontier AI companies in the Bay Area last month. On a lighter note - I have never felt poorer in my life, and being an electrical engineer by education regretted my career choices as much. For those who haven't been to the region recently, 9 out of 10 hoardings in the city are AI related, [Exhibit 1](#) shows pictures I took (as compared to only real-estate hoardings in the city I live, Mumbai), I never thought I would ever see 'MWs' on a billboard. In fact some houses in bay-areas are up for sale not for [cash but stocks of AI Companies](#). My key takeaways from discussions (not my views):

a) *Is there an AI bubble?* Is AI revolutionary - absolutely 'Yes', but in stocks almost all agree a reality check is needed. Most employees were diversifying away from their company stocks. The debt fallout could be worse, and have broader implications some felt- as incrementally questions get asked on returns on capex. b) *The long term role of AI?* The best explanation I got was it's a bit like when the internet was gaining traction in 1990s, initially a bubble was built as the world thought everything will move to the net; while the bubble burst in 2000-02, over time internet became indispensable and we had many large winners. c) *The impact of Chinese models?* One of the starkest examples I got was a company which was earlier using one of the frontier American models only to switch to a Chinese open-source model and using their American partner only for computing power - effectively reducing their cost to a fraction of earlier. But you still need compute.

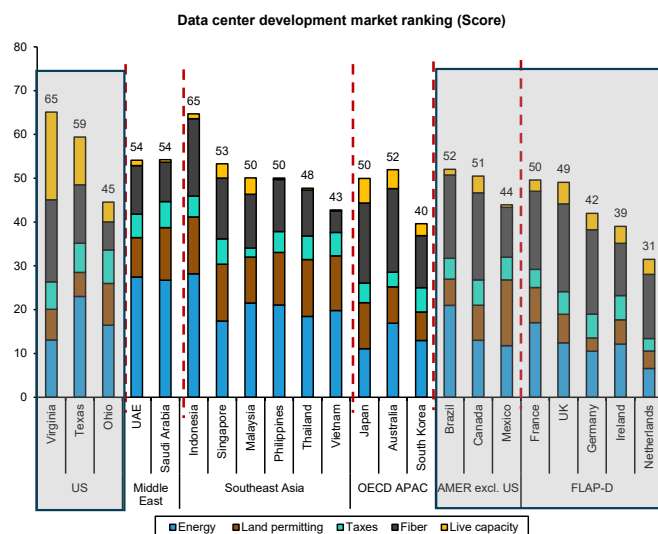
EXHIBIT 1: Billboards in Bay-area are all about AI...(pics from July'26)



Source: Bernstein analysis

IS THERE A CASE FOR ASIA IN THE DATA-CENTER SURGE?

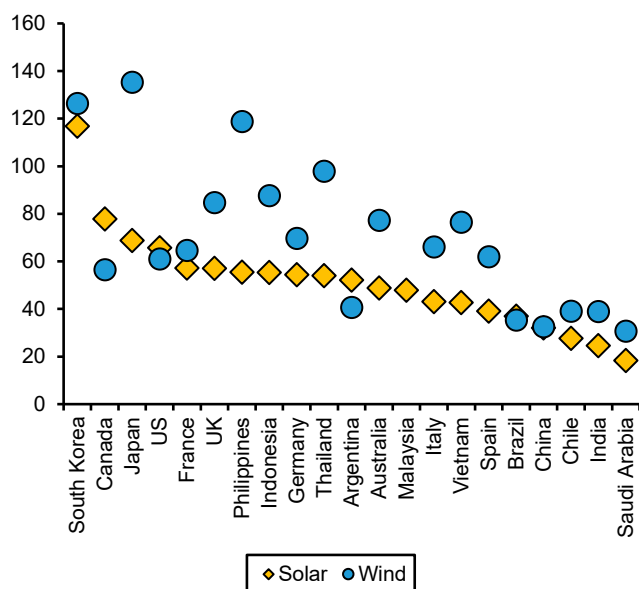
The main parameters in the decision for location of DCs are electricity availability (and cost), land, fiber network, tax breaks and existing DC ecosystem in the region. Hence, by nature US wins given its large established DC set-up, fiber network and customer base - but now gradually grid-land (not in my backyard) constraints in US hot-spots are increasing attraction of markets beyond US as well. If we look at the Bloomberg DC attractiveness rankings, Asian markets rank quite highly on most parameters. While the developed countries like Japan-Singapore rank well due to their established DC capacity and fiber connectivity, emerging markets in Middle-East, Indonesia, Malaysia seem very well-placed due to access to electricity and land. While geopolitical factors could impact growth in M.East and some other markets near term - the long term advantages of these markets are hard to dismiss. India while not in the rankings makes for solid case as well.

EXHIBIT 2: Data center development market ranking by Bloomberg


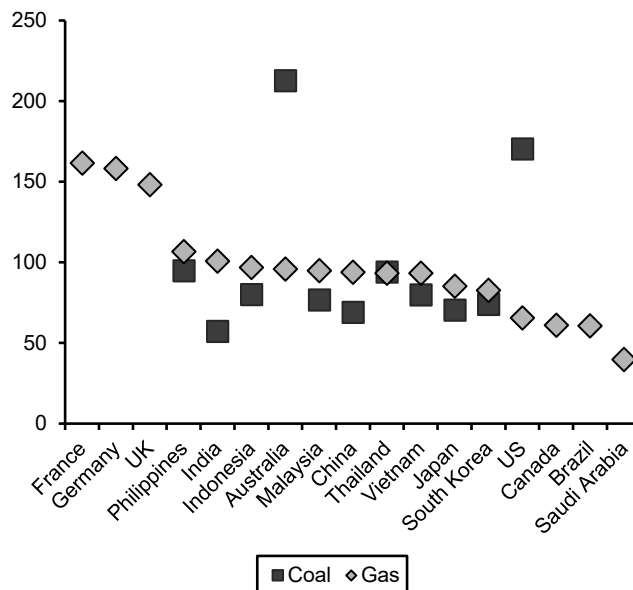
Source: Bloomberg, Bernstein analysis

1) LOW COST POWER...

The biggest reason for emerging Asia to stand out is access to electricity, this is visible in the charts below where in levelised cost of renewable power India-China-M.East are lowest in the world, and even in conventional power costs for many Asian markets are very competitive (especially coal).

EXHIBIT 3: Solar and wind LCOE comparison
2025 Solar and wind LCOE (\$/MWh)


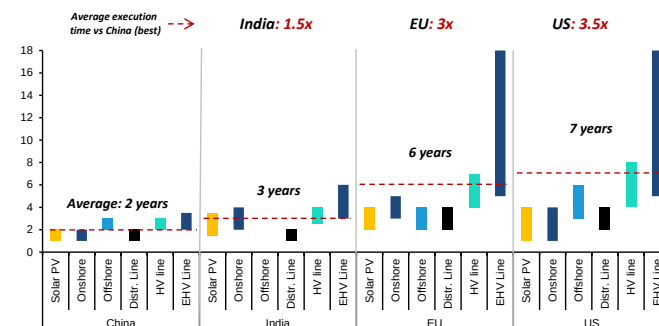
Source: Bloomberg, Bernstein analysis

EXHIBIT 4: Coal and gas LCOE comparison for select countries
2025 Coal and gas (CCGT) LCOE (\$/MWh)


Source: Bloomberg, Bernstein analysis

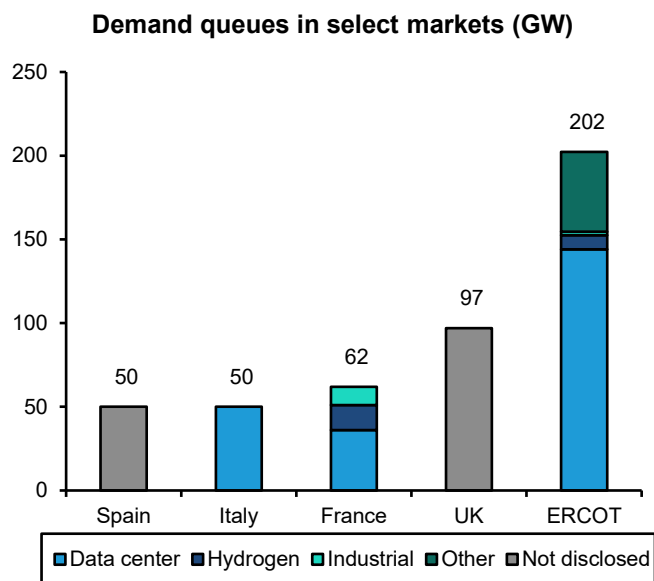
2) THEY BUILD TRANSMISSION GRIDS MUCH FASTER...

The other factor in support of emerging Asian markets is the timelines for grid connectivity. Most developed markets including the US have very long demand-supply queues, waiting for transmission connectivity, which take many years to build in these countries - given extensive timelines to get approvals. Whereas developing markets like India can build HV lines in 2-3 years.

EXHIBIT 5: Grid buildout timelines: China-India build significantly faster than developed countries


Source: IEA, Bernstein analysis

EXHIBIT 6: Large global interconnection queue on the demand side



Data for Spain is as of September 2025, France December 2024, UK June 2025, Ercot as of September 2025

Source: Bloomberg, Bernstein analysis

3) ASIAN MARKETS OFFER A BIG CUSTOMER BASE...

India is the 2nd biggest market for both Claude and Chatgpt in terms of usage, just by its sheer population size. Other Asian markets also form a sizable base of customers for AI tools, strengthening the case for the rise in DC capacity in these markets.

EXHIBIT 7: Claude usage by country (reported Sep 2025)

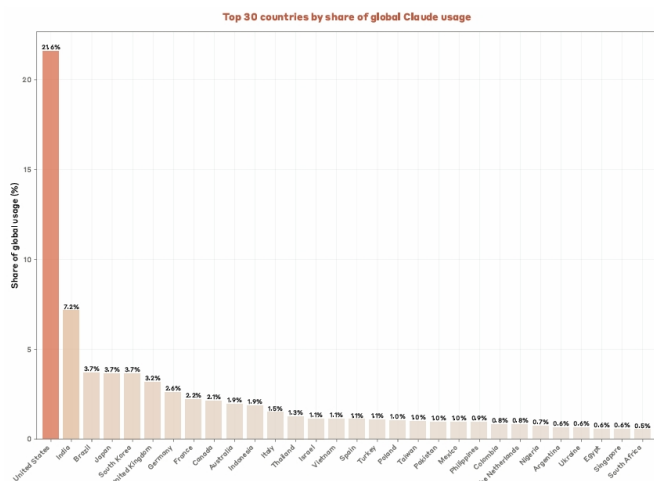


Figure 2.1: Leading countries in terms of global Claude.ai usage share The data includes Claude.ai Free and Pro conversations.

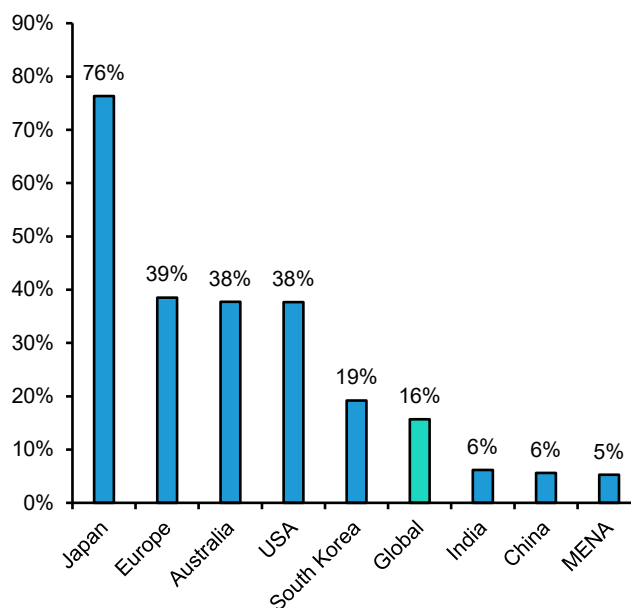
Source: Company report

IMPACT ON POWER DEMAND-SUPPLY IN MARKETS OUTSIDE THE US

Most emerging markets are prepared for high growth, hence unlike the developed markets where DC accounts for a massive share of incremental demand, for emerging markets like India despite a strong DC story its contribution is expected to be just 5-6% of incremental demand. Hence, they can easily absorb such power demand surprises.

EXHIBIT 8: Power demand for DC share in incremental power demand in select markets

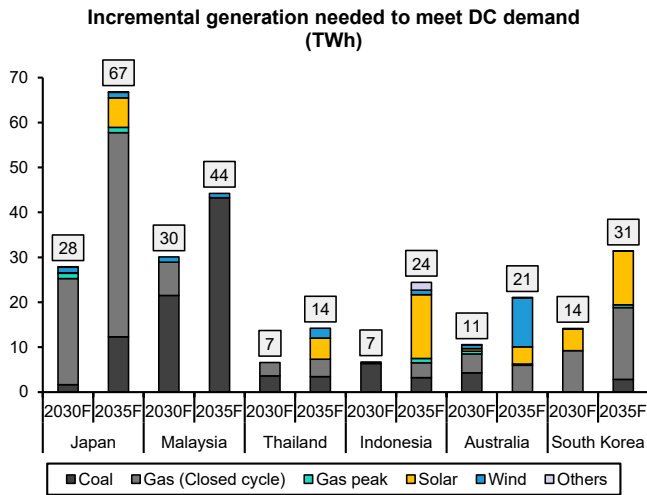
Share of data centers in incremental power demand (2025-30F) (%)



Source: Bloomberg, Bernstein analysis

Incremental DC demand is expected to be met via gas in developed Asia and in most emerging ones via coal based generation. Only post 2030, is the share of renewables to power up DCs expected to go up.

EXHIBIT 9: Incremental generation required to meet DC demand for select countries mainly through coal and gas



Source: Bloomberg, Bernstein analysis

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	5 Aug 2026	Price Target	TTM	Reported EPS				Reported P/E (x)		
			Closing Price		Rel. Perf.	Cur	2026A	2027E	2028E	2026A	2027E	2028E
LT.IN (L&T)	O	INR	4,057.00	4,637.00	(17.9)%	INR	109.34	123.40	154.92	23.5	20.8	17.2
NTPC.IN (NTPC Ltd)	O	INR	349.90	450.00	(25.9)%	INR	27.90	27.06	28.42	12.5	12.9	12.3
PWGR.IN (Power Grid)	O	INR	281.75	313.00	(34.1)%	INR	17.13	18.20	18.96	16.5	15.5	14.9
ADANI.IN (Adani Power)	O	INR	208.99	220.00	54.4%	INR	6.66	7.67	9.59	31.4	27.2	21.8
JSW.IN (JSW Energy)	O	INR	566.00	623.00	(24.1)%	INR	12.21	15.22	17.19	46.3	37.2	32.9
ASIAX			1,909.27									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

LT.IN valuation is EV/EBITDA (x); LT.IN base year is 2025;

Source: Bloomberg, Bernstein estimates and analysis.

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VALUATION METHODOLOGY**Larsen & Toubro Ltd**

We set a target price of Rs 4,637 using a SOTP approach. We consider their subsidiaries like LTI Mindtree, L&T technology services and L&T finance holdings with adjustments (like holding company discount) to market cap and value non-services business using forward P/E ratio of 30x on our FY28 EPS estimates.

NTPC Ltd

We value NTPC using DCF method to arrive at our INR450 price target. On the conventional business, we forecast the financials till 2070 asset-wise - assuming no further thermal/nuclear capacity addition. For renewables, we add NTPC Green's valuation with a 30% holdco discount. We have used a weighted average cost of capital (WACC) of 8%.

Power Grid Corp of India Ltd

We value Power Grid using DCF methodology to arrive at our INR 313 price target. We apply cost of equity of 9.5% and pre-tax cost of debt of 7%. We model full cash flows for assets to be commissioned pre 2032, post which we take a steady annual capex for new assets and assume WACC+2% return on them.

Adani Power

We value Adani Power using DCF method to arrive at our INR220 price target. We assume 24GW pipeline to get added by FY33 and forecast the financials asset wise till assumed 40 year life. Post FY33, we have assumed 1.5GW/yr addition till FY60. We have used a weighted average cost of capital (WACC) of 10%.

JSW Energy

We value JSW Energy using DCF method to arrive at our INR 623 price target. We add 3.2GW of coal capacity by FY32, ~3GW/yr renewable till FY30 and ~38GWh of storage (pumped storage + battery storage) by FY36 and forecast the financials asset wise. For renewables, post FY30 we separately value the renewable business assuming a ~5% market share for capacity additions till FY60. We have used a weighted average cost of capital (WACC) of 9%.

RISKS**Larsen & Toubro Ltd**

Downsides:

- 1) Prolongation of the M.East conflict
- 2) Shift in existing government stance towards infrastructure investments

- 3) Rise in commodity prices - steel, cement etc.
- 4) Rising rates impacting capex cycle
- 5) Challenges in executing projects in Middle-East due to local sourcing requirements or working capital stress
- 6) Slowdown in Middle East capex

NTPC Ltd

Downside:

1. Power demand in India growing slower than our anticipation of 5-6% CAGR
2. Uncertainty on ordering of thermal plants beyond the under-construction capacity, given competition from private sector
3. Further disappointment in renewable capacity addition by its subsidiary
4. Nuclear - No traction in nuclear project award

Power Grid Corp of India Ltd

Downside risks:

1. Slower than expected project capitalization due to right of way issues
2. Slower transmission project tendering as govt. looks to optimize capex
3. Equipment availability becoming a bottleneck for transmission capacity addition given the global requirement for transmission equipment.
4. New Opportunities - Slower tendering in intra-state or absence in battery opportunities.

Adani Power

Downside:

1. Rapid reduction in battery prices reducing the relevance of adding a lot thermal capacity
2. Negotiation with Bangladesh government on Godda plant tariff not going in the favor of Adani Power
3. Power demand in India growing slower than our anticipation of 5-6% CAGR
4. Risk of re-negotiation of long term contracts being signed for thermal power plants

JSW Energy

Downside:

1. Power demand in India growing slower than our anticipation of 5-6% CAGR
2. Risk of re-negotiation of long term contracts being signed for thermal power plants and possibility of delays in thermal projects
3. Renewable and storage addition being slower than anticipated due to land and transmission challenges
4. High leverage and a tough equity market making it challenging for the company to raise additional financing

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

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Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

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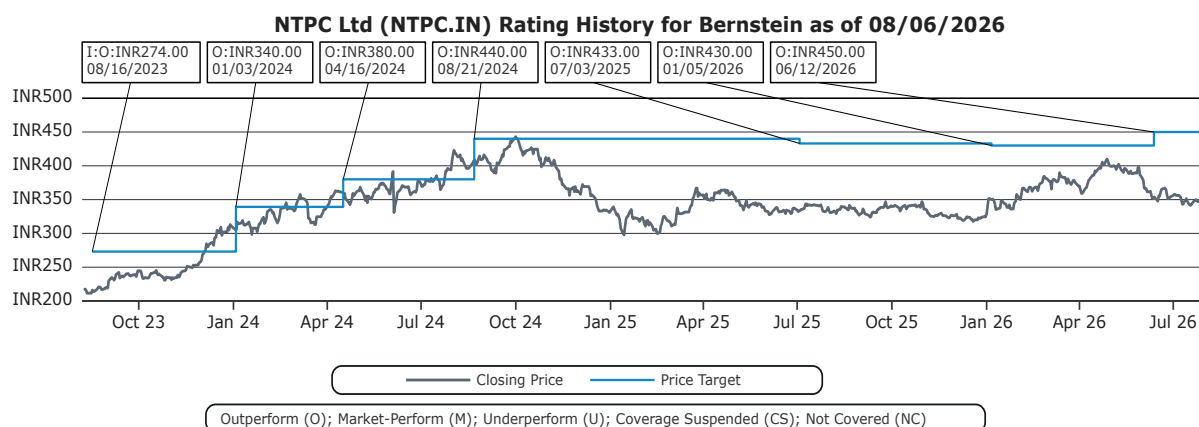
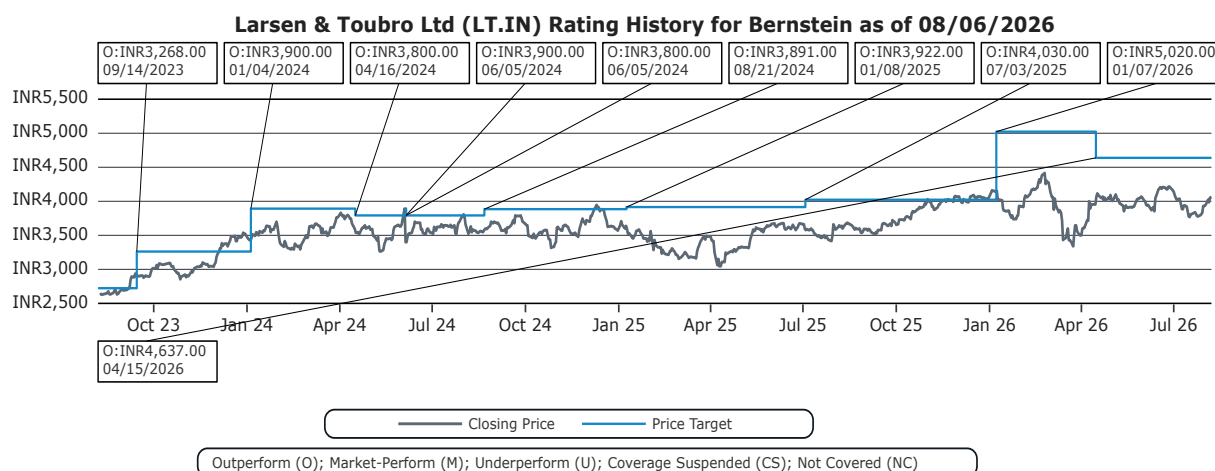
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Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of June 30, 2026. All figures are updated quarterly.

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